Conditional Statements — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Hypothesis and Conclusion

Watch Me — Worked Example

"If two angles are vertical angles, then they are congruent."
Hypothesis (p): two angles are vertical angles. Conclusion (q): they are congruent.
Not in if-then form? "All squares are rectangles" $\rightarrow$ If a shape is a square, then it is a rectangle.

1. "If $x = 5$, then $2x = 10$." Name the hypothesis.

$x = 5$

2. "If a figure is a triangle, then it has three sides." Name the conclusion.

it has three sides

3. Write in if-then form: "All right angles measure $90°$."

If an angle is right, then it measures $90°$.

4. Write in if-then form: "Fish live in water."

If an animal is a fish, then it lives in water.

Type 2 Converse, Inverse, Contrapositive

Watch Me — Worked Example

Conditional: If an animal is a dog, then it has four legs.
Converse (switch): If an animal has four legs, then it is a dog.
Inverse (negate): If an animal is not a dog, then it does not have four legs.
Contrapositive (switch and negate): If an animal does not have four legs, then it is not a dog.

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5. "If you live in Los Angeles, then you live in California." Write the converse.

If you live in California, then you live in Los Angeles.

6. "If an angle measures $120°$, then it is obtuse." Write the contrapositive.

If an angle is not obtuse, then it does not measure $120°$.

7. "If $x = 3$, then $x^2 = 9$." Write the inverse.

If $x \neq 3$, then $x^2 \neq 9$.

8. p: it is Saturday. q: there is no school. Write $\sim q \rightarrow \sim p$ in words.

If there is school, then it is not Saturday.

Type 3 True or False? Give a Counterexample

Watch Me — Worked Example

"If $x^2 = 9$, then $x = 3$." Try $x = -3$: $(-3)^2 = 9$, but $x \neq 3$.
Answer: false — counterexample $x = -3$. (To show a statement is false, one counterexample is enough.)

9. "If a number is divisible by 10, then it is divisible by 5." True or false?

true

10. Converse of #9: "If a number is divisible by 5, then it is divisible by 10." True or false? Counterexample?

false; 15

11. "If two angles are supplementary, then they form a linear pair." True or false?

false (they need not be adjacent)

12. A conditional is true. Must its contrapositive be true? Must its converse be true?

contrapositive yes; converse no

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Type 4 Biconditionals

Watch Me — Worked Example

"If an angle is a right angle, then it measures $90°$" is true, and so is its converse.
Biconditional: An angle is a right angle **if and only if** it measures $90°$.

13. Write as a biconditional: "If two angles have the same measure, then they are congruent" (the converse is also true).

Two angles are congruent if and only if they have the same measure.

14. Is this a good biconditional? "An angle is acute if and only if it measures $40°$." Explain.

No: a $60°$ angle is acute but is not $40°$.

15. Write the two conditionals in: "$x + 2 = 7$ if and only if $x = 5$."

If $x + 2 = 7$, then $x = 5$; if $x = 5$, then $x + 2 = 7$.

16. HARD A statement is false. Which related statement must also be false: converse, inverse or contrapositive?

the contrapositive

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